Youyang Shang

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EDUCATION

The Ohio State University — Columbus, OH
Bachelor of Science in Mathematics(Honors theoretical track) — August 2026 - December 2029 (Expected)
Minor in Computer Science — August 2026 - December 2029 (Expected)

TECHNICAL SKILLS

Programming: Familar in C++, C#, Java and Unity Engine Editor. Understand object-oriented programming (OOP). Known JavaScript, Python, basic data structure, algorithm design, and some Quantum Cryptography.
Mathematics: Mathematical Analysis; Statistics
Languages: Mandarin Chinese (Native); English (Fluent)

RESEARCH EXPERIENCE

Math Cycle Program — *September 2026 - Present*
Topic: Fractal Geometry
Mentor: Ori Salim Friesen (Ph.D. Student in Mathematics)

- Studies geometric patterns with self-similarity and complex structures, including fractal dimensions, recursive generation, and complex forms found in nature.
- Uses computational simulations and mathematical proofs to analyze fractal models, investigate their properties, and establish their underlying mathematical behavior.

LEADERSHIP

Buckeyes ACG Piano Club — The Ohio State University
Founder and President — *Starting February 2027 (Expected)*

- Founded and lead a student organization that brings together piano enthusiasts and ACG (Anime, Comics, and Games) fans through music and shared interests.
- Organize piano performances, music theory tutoring sessions, and collaborative workshops while managing club operations, recruiting members, and fostering an inclusive creative community.

PROJECTS & EXTRA CURRICULUMS

CampusPilot - HackOHI/O 2026 — Hackathon Prototype and Ongoing Development
Team Leader, Algorithm Designer — *October 2026 - Present*

- Developed a campus task-planning application with three teammates using React, TypeScript, and Python.
- Tested scheduling algorithms and verified application functionality through automated test cases.
- Post-hackathon: developing quantum optimization extensions using IBM Qiskit; room-booking functionality remains in development.

Dimensional Adventure - Autumn GameJam 2026 — Held by OSU Game Creation Club
Programmer, Prefabs, Assets and UI designer — *September 2026*

- Contributed to the development of Dimensional Adventure, a Unity 2D pixel-art GameJam project, focusing on gameplay programming, prefab creation and management, and asset integration.
- Designed and implemented the game's user interface, including the main menu and in-game UI elements, to support intuitive navigation and a cohesive player experience.

Quantum Cryptograph - Qiskit Fall Fest 2026 — Hackathon Prototype and Ongoing Development
Programmer, Prefabs, Assets and UI designer — *November 2026*

- Participated in the Qiskit Fall Fest Quantum Cryptography Challenge, a problem-solving competition involving quantum algorithms to recover encryption keys and decrypt 10 locked files.
- Developed post-processing and decryption workflows to convert quantum algorithm outputs into valid encryption keys.
- Optimized quantum circuits, independently verified solutions through official tests and randomized trials, and managed final code packaging and submission.

AstralNote - ACG Music Band Help by OSU JPAS Club

Intern Keyboarder *November 2026 - Present*

- Performed keyboard accompaniment for anime and ACG music band rehearsals and live performances.
- Assisted with musical arrangements, adapted to different styles, and collaborated with band members to ensure cohesive performances.